

AI-ML Assignment – 3

Total Marks: 10

Topic

Salary Prediction using Polynomial Regression



Submission Deadline

23 July 2026, 11:59 PM IST



Assignment PDF

Download the assignment from the shared Google Drive folder:

Google Drive Folder

https://drive.google.com/drive/folders/14_OGcBqYcEHc0F8Zfk_ezeqgf7LFcoDgF?usp=drive_link



Dataset

Position Salaries Dataset

Kaggle Link:

<https://www.kaggle.com/datasets/akram24/position-salaries>

Problem Statement

A company wants to estimate the **salary of employees** based on their **position level**. Since the relationship between position level and salary is non-linear, develop a **Polynomial Regression** model to predict employee salaries.

Assignment Tasks (Total: 10 Marks)

Task 1: Data Understanding (2 Marks)

1. Load the dataset using Pandas.
 2. Display the first five records.
 3. Identify:
 - Input Feature
 - Target Variable
 4. Display the dataset information and summary statistics.
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Task 2: Data Preprocessing (2 Marks)

Perform the following:

- Check for missing values.
 - Select the appropriate feature(s) and target variable.
 - Split the dataset into **80% training** and **20% testing**.
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Task 3: Model Development (3 Marks)

1. Transform the input feature using **Polynomial Features (Degree = 3)**.
 2. Train a **Polynomial Regression** model.
 3. Predict salaries for the test dataset.
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Task 4: Model Evaluation (2 Marks)

Evaluate the model using:

- Mean Absolute Error (MAE)
- Mean Squared Error (MSE)
- R^2 Score

Also create:

- Scatter plot of the original data
- Polynomial Regression Curve

Write **2–3 observations** based on the model performance.

Task 5: Conclusion (1 Mark)

Write a **100–150 word** conclusion covering:

- Key findings
 - Difference between Linear Regression and Polynomial Regression
 - One advantage of Polynomial Regression for this dataset
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Submission Instructions

Create a **public GitHub repository** containing:

- `Assignment-3.ipynb` or `Assignment-3.py`
- `README.md` including:
 - Objective
 - Dataset Link
 - Libraries Used
 - Methodology
 - Results
 - Conclusion

Do not upload the dataset to GitHub unless its license explicitly allows redistribution. Instead, provide the Kaggle dataset link in your README.md.

Google Form Submission

Submit your assignment using the Google Form:

<https://forms.gle/fFL2CFooc5Vb2MXq8>

Fill in:

- Name
- Registration Number
- Application Number
- Batch Number
- **Assignment Number (Assignment -3)**
- Public GitHub Repository Link
- Email Address

Important Instructions

- **Deadline: 23 July 2026, 11:59 PM IST**
- Only the **first valid submission** will be accepted.
- Late submissions will be automatically rejected.
- Duplicate submissions will be automatically rejected.
- Future assignment submissions are not permitted.
- Ensure your GitHub repository remains **public** until evaluation is completed.
- You will receive an email confirming whether your submission has been **accepted** or **rejected**.